

Newman-Penrose Formalism Calculations

Stephani et al., Equation (25.59)

Metric

$$ds^2 = \left(\frac{2(t^2 + z^2)}{at^2 - 2bt - 2c} \right) dt \otimes dt + \left(-\frac{at^4 z^2 - 2mt^4 z - 2nt^4 + (at^2 - 2bt - 2c)z^4}{2(t^2 + z^2)} \right) dx \otimes dx + \left(\frac{mt^2 z + nt^2 - (bt + c)z^2}{t^2 + z^2} \right) dx \otimes dy \\ + \left(\frac{mt^2 z + nt^2 - (bt + c)z^2}{t^2 + z^2} \right) dy \otimes dx + \left(-\frac{at^2 + az^2 - 2bt - 2mz - 2c - 2n}{2(t^2 + z^2)} \right) dy \otimes dy + \left(-\frac{2(t^2 + z^2)}{az^2 - 2mz - 2n} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{\frac{1}{2}t^2 + \frac{1}{2}z^2}}{\sqrt{-\frac{1}{2}at^2 + bt + c}} \right) dt + \left(\frac{\sqrt{\frac{1}{2}t^2 + \frac{1}{2}z^2}}{\sqrt{-\frac{1}{2}az^2 + mz + n}} \right) dz$$

$$n = \left(\frac{\sqrt{\frac{1}{2}t^2 + \frac{1}{2}z^2}}{\sqrt{-\frac{1}{2}at^2 + bt + c}} \right) dt + \left(-\frac{\sqrt{\frac{1}{2}t^2 + \frac{1}{2}z^2}}{\sqrt{-\frac{1}{2}az^2 + mz + n}} \right) dz$$

$$m = \left(\frac{i\sqrt{-\frac{1}{2}az^2 + mz + nt^2}}{\sqrt{2t^2 + 2z^2}} - \frac{\sqrt{-\frac{1}{2}at^2 + bt + cz^2}}{\sqrt{2t^2 + 2z^2}} \right) dx + \left(\frac{\sqrt{-\frac{1}{2}at^2 + bt + c}}{\sqrt{2t^2 + 2z^2}} + \frac{i\sqrt{-\frac{1}{2}az^2 + mz + n}}{\sqrt{2t^2 + 2z^2}} \right) dy$$

$$\bar{m} = \left(-\frac{i\sqrt{-\frac{1}{2}az^2 + mz + nt^2}}{\sqrt{2t^2 + 2z^2}} - \frac{\sqrt{-\frac{1}{2}at^2 + bt + cz^2}}{\sqrt{2t^2 + 2z^2}} \right) dx + \left(\frac{\sqrt{-\frac{1}{2}at^2 + bt + c}}{\sqrt{2t^2 + 2z^2}} - \frac{i\sqrt{-\frac{1}{2}az^2 + mz + n}}{\sqrt{2t^2 + 2z^2}} \right) dy$$

Contravariant components

$$l = \left(\frac{at^2}{2\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}} - \frac{bt}{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}} - \frac{c}{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}} \right) dt + \left(\frac{\sqrt{-az^2 + 2mz + 2n}}{2\sqrt{t^2 + z^2}} \right) dz$$

$$n = \left(\frac{at^2}{2\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}} - \frac{bt}{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}} - \frac{c}{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}} \right) dt + \left(\frac{az^2}{2\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}} - \frac{mz}{\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}} - \frac{n}{\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}} \right) dz$$

$$m = \left(-\frac{i\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}at^2}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} + \frac{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}az^2}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} + \frac{2i\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}bt}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} - \frac{2\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}mz}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} + \frac{2i\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}c}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} - \frac{2\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}n}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \right) dx + \left(-\frac{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}at^2z^2}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} - \frac{i\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}at^2z^2}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} + \frac{2\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}mt^2z}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} + \frac{2i\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}btz^2}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} + \frac{2\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}}nt^2}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} + \frac{2i\sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}}cz^2}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \right) dy$$

$$\begin{aligned}
& \overline{m} \\
& = \left(\frac{i \sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2} at^2}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \right. \\
& + \frac{\sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2} az^2}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{2i \sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2} bt}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{2 \sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2} mz}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{2i \sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2} c}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& \left. - \frac{2 \sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2} n}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \right) dx \\
& + \left(- \frac{\sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2} at^2 z^2}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \right. \\
& + \frac{i \sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2} at^2 z^2}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{2 \sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2} mt^2 z}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{2i \sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2} btz^2}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{2 \sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2} nt^2}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& \left. - \frac{2i \sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2} cz^2}{a^2 t^4 z^2 + a^2 t^2 z^4 - 2amt^4 z - 2abt^3 z^2 - 2amt^2 z^3 - 2abtz^4 - 2ant^4 + 4bmt^3 z - 2act^2 z^2 - 2ant^2 z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2 z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \right) dy
\end{aligned}$$

Directional Derivatives

$$D = \frac{at^2}{2 \sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2}} - \frac{bt}{\sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2}} - \frac{c}{\sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2}} \partial_t + \frac{\sqrt{-az^2 + 2mz + 2n}}{2 \sqrt{t^2 + z^2}} \partial_z$$

$$\begin{aligned}
\Delta = & \frac{at^2}{2 \sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2}} - \frac{bt}{\sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2}} - \frac{c}{\sqrt{-at^2 + 2bt + 2c} \sqrt{t^2 + z^2}} \partial_t \\
& + \frac{az^2}{2 \sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2}} - \frac{mz}{\sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2}} - \frac{n}{\sqrt{-az^2 + 2mz + 2n} \sqrt{t^2 + z^2}} \partial_z
\end{aligned}$$

$\delta =$

$$\begin{aligned}
& - \frac{i \sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}at^2}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}az^2}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{2i \sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}bt}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{2 \sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}mz}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{2i \sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}c}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{2 \sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}n}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \partial_x \\
& - \frac{\sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}at^2z^2}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{i \sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}at^2z^2}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& - \frac{2 \sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}mt^2z}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{2i \sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}btz^2}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{2 \sqrt{-at^2 + 2bt + 2c\sqrt{t^2 + z^2}nt^2}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \\
& + \frac{2i \sqrt{-az^2 + 2mz + 2n\sqrt{t^2 + z^2}cz^2}}{a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2} \partial_y
\end{aligned}$$

$\bar{\delta}$

$$\begin{aligned}
&= \frac{i\sqrt{-az^2+2mz+2n}\sqrt{t^2+z^2}at^2}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&+ \frac{\sqrt{-at^2+2bt+2c}\sqrt{t^2+z^2}az^2}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&- \frac{2i\sqrt{-az^2+2mz+2n}\sqrt{t^2+z^2}bt}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&- \frac{2\sqrt{-at^2+2bt+2c}\sqrt{t^2+z^2}mz}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&- \frac{2i\sqrt{-az^2+2mz+2n}\sqrt{t^2+z^2}c}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&- \frac{2\sqrt{-at^2+2bt+2c}\sqrt{t^2+z^2}n}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \partial_x \\
&- \frac{\sqrt{-at^2+2bt+2c}\sqrt{t^2+z^2}at^2z^2}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&+ \frac{i\sqrt{-az^2+2mz+2n}\sqrt{t^2+z^2}at^2z^2}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&+ \frac{2\sqrt{-at^2+2bt+2c}\sqrt{t^2+z^2}mt^2z}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&- \frac{2i\sqrt{-az^2+2mz+2n}\sqrt{t^2+z^2}btz^2}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&- \frac{2\sqrt{-at^2+2bt+2c}\sqrt{t^2+z^2}nt^2}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \\
&+ \frac{2i\sqrt{-az^2+2mz+2n}\sqrt{t^2+z^2}cz^2}{a^2t^4z^2+a^2t^2z^4-2amt^4z-2abt^3z^2-2amt^2z^3-2abtz^4-2ant^4+4bmt^3z-2act^2z^2-2ant^2z^2+4bmtz^3-2acz^4+4bnt^3+4cmt^2z+4bntz^2+4cmz^3+4cnt^2+4cnz^2} \partial_y
\end{aligned}$$

Spin Coefficients

$\rho =$

$$\begin{aligned}
& - \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}a^2t^2z}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}a^2tz^2}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}amt^2}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}abtz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}amtz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}abz^2}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}bmt}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}ant}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}acz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}bmz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}cm}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}bn}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)}
\end{aligned}$$

[illegible]

$$\kappa = 0$$

$$\mu =$$

$$\begin{aligned}
& - \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}a^2t^2z}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}a^2tz^2}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}amt^2}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}abtz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}amtz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}abz^2}{4(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}bmt}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}ant}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& + \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}acz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}bmz}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}cm}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)} \\
& - \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}bn}{2(a^2t^4z^2 + a^2t^2z^4 - 2amt^4z - 2abt^3z^2 - 2amt^2z^3 - 2abtz^4 - 2ant^4 + 4bmt^3z - 2act^2z^2 - 2ant^2z^2 + 4bmtz^3 - 2acz^4 + 4bnt^3 + 4cmt^2z + 4bntz^2 + 4cmz^3 + 4cnt^2 + 4cnz^2)}
\end{aligned}$$

[illegible]

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}t}{4(t^4 + 2t^2z^2 + z^4)} + \frac{i\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}t}{4(t^4 + 2t^2z^2 + z^4)} + \frac{i\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}z}{4(t^4 + 2t^2z^2 + z^4)} + \frac{\sqrt{-az^2 + 2mz + 2n}\sqrt{t^2 + z^2}z}{4(t^4 + 2t^2z^2 + z^4)}$$

$$\begin{aligned} \gamma = & -\frac{i\sqrt{t^2 + z^2}atz^2}{4(t^4 + 2t^2z^2 + z^4)\sqrt{-az^2 + 2mz + 2n}} - \frac{\sqrt{t^2 + z^2}az^3}{4(t^4 + 2t^2z^2 + z^4)\sqrt{-az^2 + 2mz + 2n}} + \frac{i\sqrt{t^2 + z^2}mtz}{2(t^4 + 2t^2z^2 + z^4)\sqrt{-az^2 + 2mz + 2n}} + \frac{\sqrt{t^2 + z^2}mz^2}{2(t^4 + 2t^2z^2 + z^4)\sqrt{-az^2 + 2mz + 2n}} \\ & + \frac{\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}t}{4(t^4 + 2t^2z^2 + z^4)} + \frac{i\sqrt{t^2 + z^2}nt}{2(t^4 + 2t^2z^2 + z^4)\sqrt{-az^2 + 2mz + 2n}} - \frac{i\sqrt{-at^2 + 2bt + 2c}\sqrt{t^2 + z^2}z}{4(t^4 + 2t^2z^2 + z^4)} + \frac{\sqrt{t^2 + z^2}nz}{2(t^4 + 2t^2z^2 + z^4)\sqrt{-az^2 + 2mz + 2n}} \end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{c}{2(t^4 + 2t^2z^2 + z^4)} - \frac{n}{2(t^4 + 2t^2z^2 + z^4)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{c}{2(t^4 + 2t^2z^2 + z^4)} + \frac{n}{2(t^4 + 2t^2z^2 + z^4)}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{c}{2(t^4 + 2t^2z^2 + z^4)} + \frac{n}{2(t^4 + 2t^2z^2 + z^4)}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{c}{2(t^4 + 2t^2z^2 + z^4)} - \frac{n}{2(t^4 + 2t^2z^2 + z^4)}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\begin{aligned} \Psi_0 = & -\frac{3bt^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3imt^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{9ibt^2z}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{9mt^2z}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{9btz^2}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \\ & + \frac{9imtz^2}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3ibz^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{3mz^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3ct^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \\ & - \frac{3nt^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{3ictz}{t^6 + 3t^4z^2 + 3t^2z^4 + z^6} + \frac{3intz}{t^6 + 3t^4z^2 + 3t^2z^4 + z^6} + \frac{3cz^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{3nz^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned} \Psi_2 = & \frac{bt^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{imt^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3ibt^2z}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{3mt^2z}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3btz^2}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \\ & - \frac{3imtz^2}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{ibz^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{mz^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{ct^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \\ & + \frac{nt^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{ictz}{t^6 + 3t^4z^2 + 3t^2z^4 + z^6} - \frac{intz}{t^6 + 3t^4z^2 + 3t^2z^4 + z^6} - \frac{cz^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{nz^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}\Psi_4 = & -\frac{3bt^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3imt^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{9ibt^2z}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{9mt^2z}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{9btz^2}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \\ & + \frac{9imtz^2}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3ibz^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{3mz^3}{4(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} - \frac{3ct^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} \\ & - \frac{3nt^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{3ictz}{t^6 + 3t^4z^2 + 3t^2z^4 + z^6} + \frac{3intz}{t^6 + 3t^4z^2 + 3t^2z^4 + z^6} + \frac{3cz^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)} + \frac{3nz^2}{2(t^6 + 3t^4z^2 + 3t^2z^4 + z^6)}\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"